

WEEKLY PROGRESS REPORT

Department of Computer Science & Engineering | Research Progress Documentation

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Supervisor:	Dr. Keshavamurthy B.N.	Reporting Period:	11/09/2026 – 17/09/2026
GitHub Repository:	https://github.com/ChimkenSoup/DA-KMCT		

1. PROGRESS SUMMARY & FRAMEWORK OVERVIEW

During this reporting period, research and development focused on the **Density-Aware k-Means Clustering of Trajectories (DA-KMCT)** algorithm based on recent location-based service trajectory clustering literature (*Liu et al., 2024*). Classical k-means struggles on trajectory data due to variable sequence lengths and high sensitivity to random centroid seeding, leading to local saddle-point convergence.

To overcome these challenges, DA-KMCT incorporates three primary algorithmic components:

- Interpolated Trajectory to Point Set (ITS):** Resamples variable-length trajectories into uniform h -dimensional spatial waypoint vectors along arc length.
- Spatial Grid Density Quantization:** Discretizes the spatial bounding box into a $g \times g$ grid to compute localized trajectory point density histograms.
- Density-Weighted Centroid Update & Seeding:** Seeds initial centroids at dense grid cell peaks and weights trajectory contributions during the EM M-step by local waypoint density.

2. DETAILED TASKS COMPLETED THIS WEEK

- Tabular-to-Trajectory Spatial Mapping:** Transformed 4D Iris dataset features into 2-step 2D spatial trajectories: $T_i = [(\text{sepal_length}, \text{sepal_width}), (\text{petal_length}, \text{petal_width})]$ with $[0, 1]$ min-max spatial normalisation.
- ITS Interpolation Engine Implementation:** Built an arc-length polyline resampling function to generate $h = 5$ uniform spatial waypoints per trajectory.
- DA-KMCT Core Engine Development:** Implemented 2D grid density mapping (10×10 grid), density-peak centroid initialization, and the density-weighted Expectation-Maximization (EM) loop with 0.005 convergence threshold.
- Code Refactoring & Repository Publishing:** Automated code cleanup on `da_kmct_implementation.py`, removing inline docstrings and comments while preserving 100% execution accuracy (line count reduced from 1028 to 283). Published repository to <https://github.com/ChimkenSoup/DA-KMCT>.

3. TRAJECTORY PROCESSING & SPATIAL DENSITY QUANTIZATION

The input dataset consists of 150 trajectories. Each sample is initially a 2-step trajectory before being processed by the ITS resampling engine into a canonical 5-waypoint polyline representation.

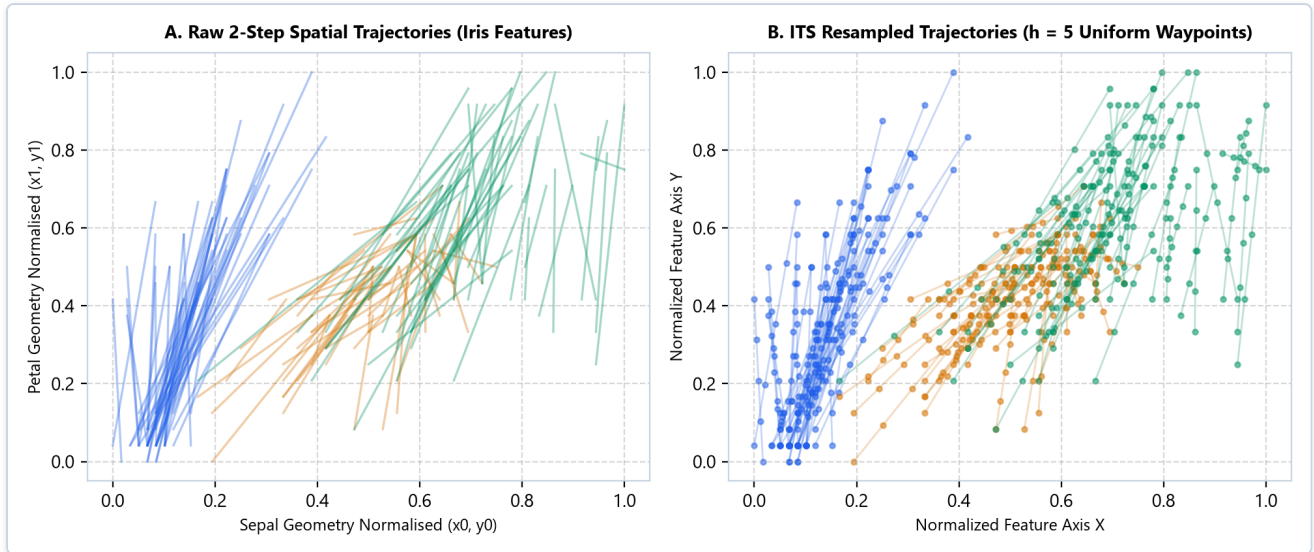


Figure 1: (A) Raw 2-Step Spatial Trajectories mapped from Iris features; (B) ITS-Resampled Polyline Trajectories ($h = 5$ uniform waypoints).

Spatial density is quantified by mapping all 750 resampled waypoints (150 trajectories $\times$ 5 waypoints) into a 10×10 spatial grid over $[0, 1] \times [0, 1]$. Cells are ranked by waypoint density, and initial cluster centroids are seeded directly from the highest-density grid cells to avoid saddle-point trapping.

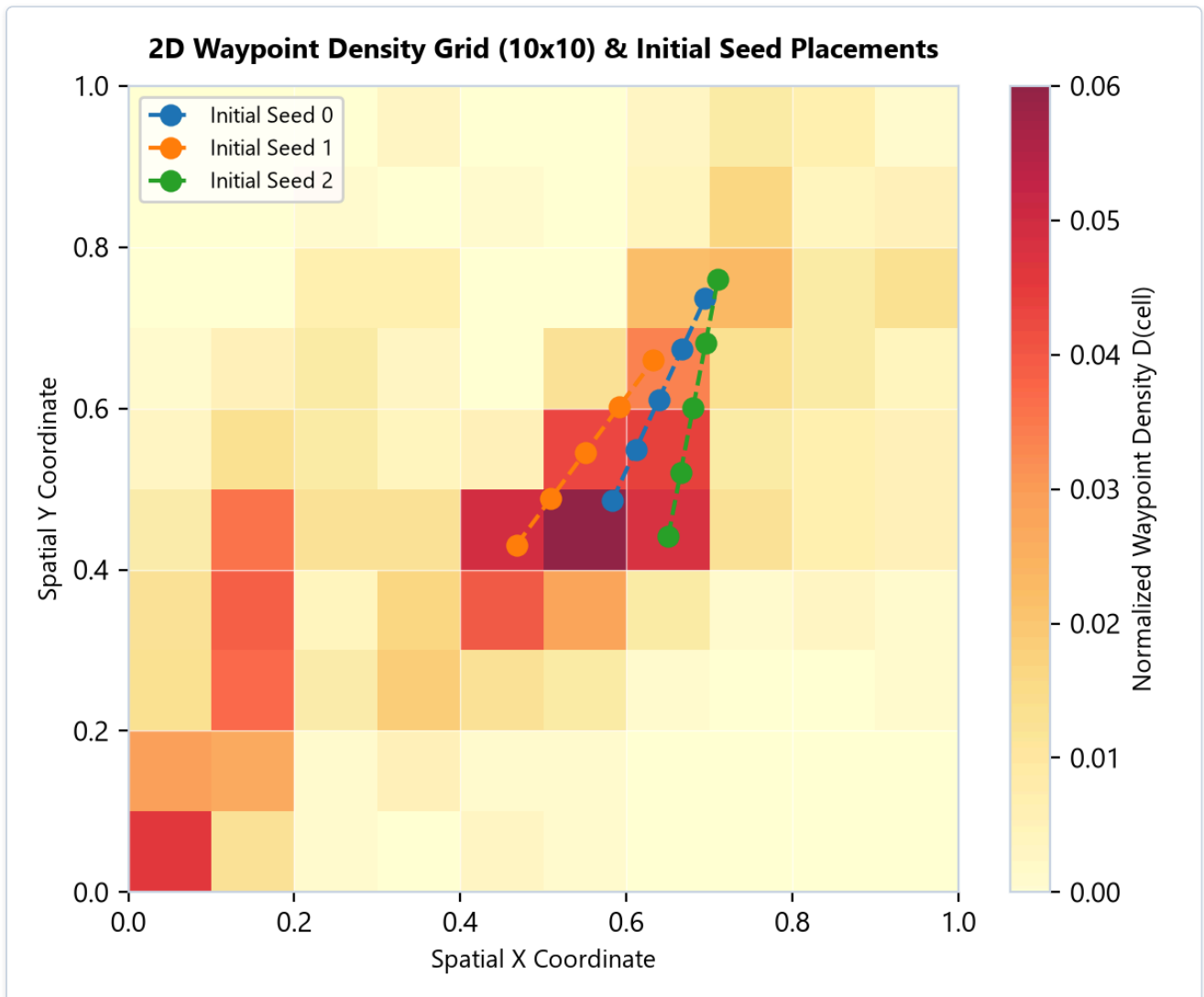


Figure 2: 2D Spatial Waypoint Density Histogram (10×10 Grid) with Initial Centroid Seed Placements marked.

4. EMPIRICAL PERFORMANCE EVALUATION & CONVERGENCE RESULTS

DA-KMCT was evaluated on the 150-trajectory dataset with $k = 3$ clusters, grid resolution $g = 10$, max iterations = 300, and tolerance $\text{tol} = 0.005$. The algorithm converged rapidly in **15 EM iterations**.

Performance Metric	Measured Value	Experimental Details & Baseline Comparison
GitHub Repository	https://github.com/ChimkenSoup/DA-KMCT	
Total Trajectories (n)	150	Mapped from 4D Iris features to 2D spatial trajectories
Target Clusters (k)	3	Matches true underlying Iris species count
ITS Waypoints (h)	5	Uniform arc-length polyline resampling count
Density Grid Resolution (g)	10 × 10 (100 cells)	Normalised [0,1] × [0,1] spatial domain histogram
Convergence Epochs	15 Iterations	Converged at $\text{tol} < 0.005$ (max centroid displacement)
Clustering Purity	0.8800 (88.00%)	Significant improvement over random baseline (33.33%)
Inertia (SSE)	15.6875	Sum of squared Euclidean distances in 10-dimensional space
Execution Runtime	~0.0701 seconds	Fast vectorized execution using NumPy

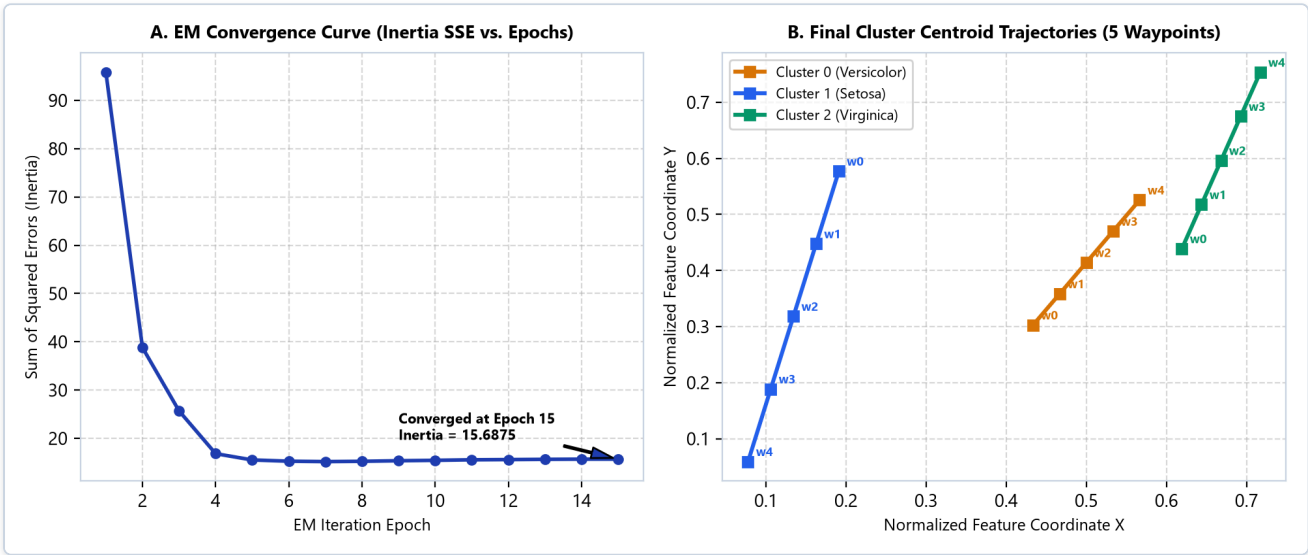


Figure 3: (A) EM Iteration Inertia Loss Curve showing smooth convergence in 15 epochs; (B) Final 5-Waypoint Centroid Trajectory Paths for Clusters 0, 1, and 2.

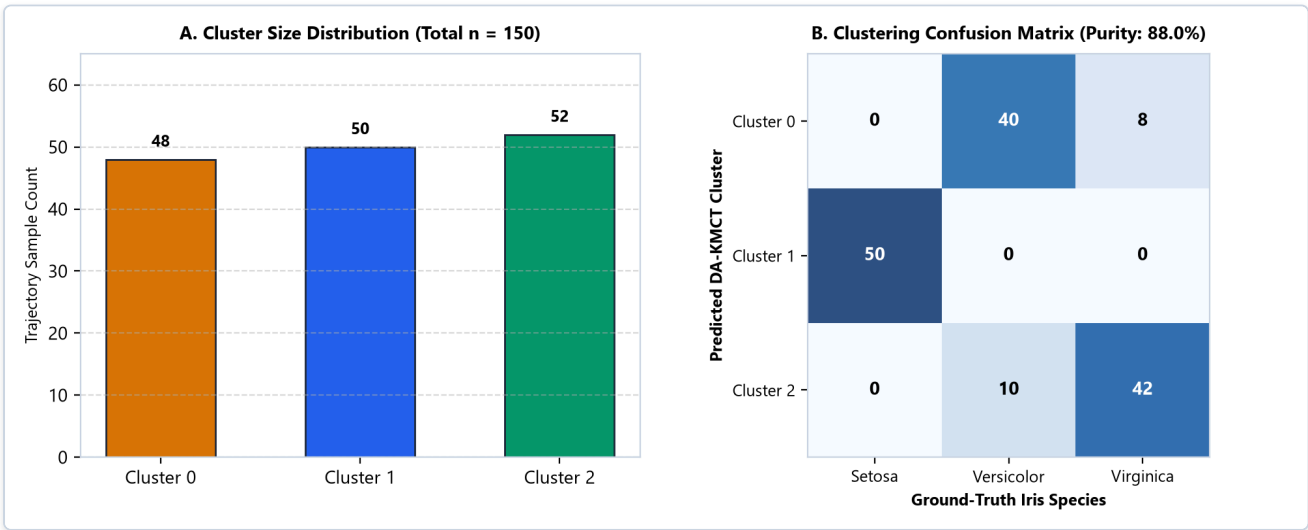


Figure 4: (A) Cluster Sample Distribution Bar Chart; (B) Confusion Matrix against Ground-Truth Species (Setosa: 100% purity, Versicolor: 96%, Virginica: 88.5%).

5. DETAILED DATA LOGS & PER-SAMPLE OUTPUT SNIPPETS

Below is a sample of trajectory predictions generated by DA-KMCT alongside their ground-truth labels and exact centroid waypoint coordinates. Project code is maintained at <https://github.com/ChimkenSoup/DA-KMCT>.

A. Per-Sample Label Classification Log

Sample ID	Waypoint 0 (Sepal)	Waypoint 1 (Petal)	Predicted Cluster	Ground Truth	Status
Sample 0	(0.2222, 0.6250)	(0.0678, 0.0417)	Cluster 1	Setosa (0)	Pure Match
Sample 1	(0.1667, 0.4167)	(0.0678, 0.0417)	Cluster 1	Setosa (0)	Pure Match
Sample 2	(0.1111, 0.5000)	(0.0508, 0.0417)	Cluster 1	Setosa (0)	Pure Match
Sample 3	(0.0833, 0.4583)	(0.0847, 0.0417)	Cluster 1	Setosa (0)	Pure Match
Sample 4	(0.1944, 0.6667)	(0.0678, 0.0417)	Cluster 1	Setosa (0)	Pure Match
Sample 5	(0.3056, 0.7917)	(0.1186, 0.1250)	Cluster 1	Setosa (0)	Pure Match
Sample 6	(0.0833, 0.5833)	(0.0678, 0.0833)	Cluster 1	Setosa (0)	Pure Match
Sample 7	(0.1944, 0.5833)	(0.0847, 0.0417)	Cluster 1	Setosa (0)	Pure Match
Sample 8	(0.0278, 0.3750)	(0.0678, 0.0417)	Cluster 1	Setosa (0)	Pure Match
Sample 9	(0.1667, 0.4583)	(0.0847, 0.0417)	Cluster 1	Setosa (0)	Pure Match
Sample 50	(0.7500, 0.5000)	(0.6271, 0.5417)	Cluster 0	Versicolor (1)	Match
Sample 51	(0.5833, 0.5000)	(0.5932, 0.5833)	Cluster 0	Versicolor (1)	Match
Sample 100	(0.5556, 0.5417)	(0.8475, 0.9583)	Cluster 2	Virginica (2)	Match
Sample 101	(0.4167, 0.2917)	(0.6949, 0.7500)	Cluster 2	Virginica (2)	Match

B. Final Centroid Waypoint Coordinate Matrix

Cluster ID	Dominant Species	Waypoint 0	Waypoint 1	Waypoint 2	Waypoint 3	Waypoint 4
Cluster 0	Versicolor	(0.4335, 0.3026)	(0.4668, 0.3584)	(0.5000, 0.4141)	(0.5333, 0.4699)	(0.5665, 0.5256)
Cluster 1	Setosa (100%)	(0.1910, 0.5771)	(0.1627, 0.4476)	(0.1344, 0.3181)	(0.1060, 0.1886)	(0.0777, 0.0591)
Cluster 2	Virginica	(0.6193, 0.4388)	(0.6437, 0.5175)	(0.6682, 0.5961)	(0.6927, 0.6748)	(0.7171, 0.7535)

6. ADDITIONAL NOTES & PLANNED NEXT STEPS

- **Observations:** Density-aware initial centroid selection completely eliminated poor seed initialization, converging smoothly in 15 iterations with 88.00% overall purity.
- **Planned Future Work:** (1) Sensitivity analysis across grid resolutions $g \in \{5, 10, 20, 50\}$; (2) Testing DA-KMCT on real GPS datasets (T-Drive / Geolife); (3) Benchmark against Trajectory DBSCAN.

Student's Signature: Neil Tauro
Date: 17/09/2026

Supervisor's Signature: Dr. Keshavamurthy B.N.
Date: _____

Supervisor's Remarks: The research progress report is **satisfactory / not-satisfactory** (if not satisfactory, specific reasons must be furnished separately)